Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Systems Software Engineer with 2+ years building **C++/Python systems**, **multithreaded software**, and **automated verification** grounded in Linux, operating systems, algorithms, and debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant services** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **software verification**, validation, reconciliation, error handling, incremental processing, CI/CD releases, and stable production monitoring workflows reliably at enterprise scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship | *India*
- Reduced centralized-control and quantum-risk exposure by 25% through a **novel consensus algorithm**, comparing benchmark behavior, performance trade-offs, and failure characteristics against established blockchain mechanisms systematically and reproducibly.
- Integrated five lattice-based cryptographic primitives into a **quantum-resistant protocol**, documenting design decisions and benchmark evidence for documented peer-reviewed research presented at IISc Bengaluru through an IEEE publication.

Skills

Languages: C++, Python, Bash, Java, SQL
Systems & Architecture: Linux, Windows, Operating Systems, Memory Management, CPU Scheduling, Data Structures, Algorithms, Object-Oriented Programming, Multithreading, Concurrent Programming, Distributed Systems
Verification & Quality: Software Verification, Hardware-Software Validation, Unit Testing, Integration Testing, Automated Testing, Regression Testing, PyTest, Mockito, Debugging, Fault Injection
Cloud & Tooling: AWS, Azure, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD
Engineering Practices: Performance Optimization, Fault Tolerance, Error Handling, Monitoring, Technical Documentation, Benchmarking

Projects

Kernel Memory and Driver Verification Harness - Design Concept
- Designed a **Linux systems verification harness** concept in C++ and Python covering memory-allocation policies, concurrent workloads, simulated device I/O, deterministic fault injection, trace capture, and reproducible regression checks.

Firmware Concurrency Validation Testbench - Design Concept
- Designed a **firmware validation testbench** concept for multithreaded C++ components with mocked hardware registers, boundary-condition scenarios, race detection, structured telemetry, automated test orchestration, and hardware-software compatibility reports.

CPU-Scheduling Simulator | Live
- Built a **C++ CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios.